

# Apply filters to SQL queries

## Project description

The management at my organization has asked me to investigate potential security issues and update employee computers as required. As a Linux administrator, I used SQL with filters to perform security-related tasks.

## Retrieve after hours failed login attempts

There were suspicious activities that occurred after business hours (after 18:00). All after hours login attempts that failed need to be investigated.

I created a SQL query on MariaDB to filter for failed login attempts that occurred after business hours.

```
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE login_time > '18:00' AND success = 'FALSE';
```

| event_id | username | login_date | login_time | country | ip_address      | success |
|----------|----------|------------|------------|---------|-----------------|---------|
| 2        | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  | 0       |
| 18       | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  | 0       |
| 20       | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50  | 0       |
| 28       | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   | 0       |
| 34       | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   | 0       |
| 42       | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   | 0       |
| 52       | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57   | 0       |
| 69       | wjaffrey | 2022-05-11 | 19:55:15   | USA     | 192.168.100.17  | 0       |
| 82       | abernard | 2022-05-12 | 23:38:46   | MEX     | 192.168.234.49  | 0       |
| 87       | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 | 0       |
| 96       | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  | 0       |
| 104      | asundara | 2022-05-11 | 18:38:07   | US      | 192.168.96.200  | 0       |
| 107      | bisles   | 2022-05-12 | 20:25:57   | USA     | 192.168.116.187 | 0       |
| 111      | aestrada | 2022-05-10 | 22:00:26   | MEXICO  | 192.168.76.27   | 0       |
| 127      | abellmas | 2022-05-09 | 21:20:51   | CANADA  | 192.168.70.122  | 0       |
| 131      | bisles   | 2022-05-09 | 20:03:55   | US      | 192.168.113.171 | 0       |
| 155      | cgriffin | 2022-05-12 | 22:18:42   | USA     | 192.168.236.176 | 0       |
| 160      | jclark   | 2022-05-10 | 20:49:00   | CANADA  | 192.168.214.49  | 0       |
| 199      | yappiah  | 2022-05-11 | 19:34:48   | MEXICO  | 192.168.44.232  | 0       |

```
19 rows in set, 1 warning (0.116 sec)
```

The result is based on the `log_in_attempts` table where the `login_time` column is after 18:00 and the login attempts are failed (0). The filter “Select \* “ means to select everything (all columns) and FROM `log_in_attempts` means it is from the `log_in_attempts` table. Success indicates the status of the login. If it is zero, it is a failure whereas if it is one, it is a success. Therefore, there were 19 failed login attempts after 18:00.

## Retrieve login attempts on specific dates

A suspicious event occurred on 2022-05-09. Any login activity that happened on 2022-05-09 or on the day before needs to be investigated. Therefore, I created a SQL query to filter for login attempts that occurred on specific dates.

```
MariaDB [organization]> SELECT *  
->  
-> FROM log_in_attempts  
->  
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

| event_id | username | login_date | login_time | country | ip_address      | success |
|----------|----------|------------|------------|---------|-----------------|---------|
| 1        | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 | 1       |
| 3        | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 | 1       |
| 4        | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  | 0       |
| 8        | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 | 0       |
| 12       | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 | 1       |
| 15       | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  | 0       |
| 24       | arusso   | 2022-05-09 | 06:49:39   | MEXICO  | 192.168.171.192 | 1       |
| 25       | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  | 1       |
| 26       | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 | 1       |
| 28       | aestrada | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   | 0       |
| 30       | yappiah  | 2022-05-09 | 03:22:22   | MEX     | 192.168.124.48  | 1       |
| 32       | acook    | 2022-05-09 | 02:52:02   | CANADA  | 192.168.142.239 | 0       |
| 36       | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  | 1       |
| 38       | sbaelish | 2022-05-09 | 14:40:01   | USA     | 192.168.60.42   | 1       |
| 39       | yappiah  | 2022-05-09 | 07:56:40   | MEXICO  | 192.168.57.115  | 1       |
| 42       | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   | 0       |
| 43       | mcouliba | 2022-05-08 | 02:35:34   | CANADA  | 192.168.16.208  | 0       |
| 44       | daquino  | 2022-05-08 | 07:02:35   | CANADA  | 192.168.168.144 | 0       |
| 47       | dkot     | 2022-05-08 | 05:06:45   | US      | 192.168.233.24  | 1       |
| 49       | asundara | 2022-05-08 | 14:00:01   | US      | 192.168.173.213 | 0       |
| 53       | nmason   | 2022-05-08 | 11:51:38   | CAN     | 192.168.133.188 | 1       |
| 56       | acook    | 2022-05-08 | 04:56:30   | CAN     | 192.168.209.130 | 1       |
| 58       | ivelasco | 2022-05-09 | 17:20:54   | CAN     | 192.168.57.162  | 0       |
| 61       | dtanaka  | 2022-05-09 | 09:45:18   | USA     | 192.168.98.221  | 1       |
| 65       | aalonso  | 2022-05-09 | 23:42:12   | MEX     | 192.168.52.37   | 1       |
| 66       | aestrada | 2022-05-08 | 21:58:32   | MEX     | 192.168.67.223  | 1       |
| 67       | abernard | 2022-05-09 | 11:53:41   | MEX     | 192.168.118.29  | 1       |
| 68       | mrah     | 2022-05-08 | 17:16:13   | US      | 192.168.42.248  | 1       |
| 70       | tmitchel | 2022-05-09 | 10:55:17   | MEXICO  | 192.168.87.199  | 1       |
| 71       | mcouliba | 2022-05-09 | 06:57:42   | CAN     | 192.168.55.169  | 0       |
| 72       | alevitsk | 2022-05-08 | 12:09:10   | CANADA  | 192.168.139.176 | 1       |
| 79       | abernard | 2022-05-09 | 11:41:15   | MEX     | 192.168.158.170 | 0       |
| 80       | cjackson | 2022-05-08 | 02:18:10   | CANADA  | 192.168.33.140  | 1       |
| 83       | lrodriqu | 2022-05-08 | 08:10:23   | USA     | 192.168.67.69   | 1       |
| 87       | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 | 0       |
| 90       | gesparza | 2022-05-09 | 00:49:05   | CANADA  | 192.168.87.201  | 0       |
| 92       | pwashing | 2022-05-08 | 00:36:12   | US      | 192.168.247.219 | 0       |
| 96       | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  | 0       |
| 97       | irookley | 2022-05-09 | 02:40:22   | MEXICO  | 192.168.22.221  | 1       |

|     |          |            |          |        |                 |   |
|-----|----------|------------|----------|--------|-----------------|---|
| 169 | alevitsk | 2022-05-08 | 08:10:43 | CANADA | 192.168.210.228 | 0 |
| 170 | sbaelish | 2022-05-09 | 16:43:18 | USA    | 192.168.65.113  | 0 |
| 172 | mabadi   | 2022-05-08 | 08:06:50 | US     | 192.168.180.41  | 1 |
| 178 | sgilmore | 2022-05-08 | 12:27:22 | CAN    | 192.168.52.216  | 0 |
| 184 | alevitsk | 2022-05-08 | 03:09:48 | CAN    | 192.168.33.70   | 0 |
| 186 | bisles   | 2022-05-09 | 04:29:17 | USA    | 192.168.40.72   | 0 |
| 187 | arusso   | 2022-05-09 | 00:36:26 | MEX    | 192.168.77.137  | 0 |
| 189 | nmason   | 2022-05-08 | 05:37:24 | CANADA | 192.168.168.117 | 1 |
| 190 | jsoto    | 2022-05-09 | 05:09:21 | USA    | 192.168.25.60   | 0 |
| 191 | cjackson | 2022-05-08 | 06:46:07 | CANADA | 192.168.7.187   | 0 |
| 193 | lrodriqu | 2022-05-08 | 07:11:29 | US     | 192.168.125.240 | 0 |
| 197 | jsoto    | 2022-05-08 | 09:05:09 | US     | 192.168.36.21   | 0 |

75 rows in set (0.001 sec)

I selected the `log_in_attempts` table and used the `WHERE` clause and `OR` operator to filter my results to output only login attempts that occurred on 2022-05-05 or 2022-05-08. As a result, there were 75 login attempts in these two days.

## Retrieve login attempts outside of Mexico

After investigating the data and following the pattern, there is a strong indication that login attempts outside of Mexico should be investigated.

I created a SQL query to filter for login attempts that occurred outside of Mexico.

```
MariaDB [organization]> SELECT *
->
-> FROM log_in_attempts
->
-> WHERE NOT country LIKE 'MEX%';
```

| event_id | username | login_date | login_time | country | ip_address      | success |
|----------|----------|------------|------------|---------|-----------------|---------|
| 1        | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 | 1       |
| 2        | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  | 0       |
| 3        | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 | 1       |
| 4        | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  | 0       |
| 5        | jrafael  | 2022-05-11 | 03:05:59   | CANADA  | 192.168.86.232  | 0       |
| 7        | eraab    | 2022-05-11 | 01:45:14   | CAN     | 192.168.170.243 | 1       |
| 8        | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 | 0       |
| 10       | jrafael  | 2022-05-12 | 09:33:19   | CANADA  | 192.168.228.221 | 0       |
| 11       | sgilmore | 2022-05-11 | 10:16:29   | CANADA  | 192.168.140.81  | 0       |
| 12       | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 | 1       |
| 13       | mrh      | 2022-05-11 | 09:29:34   | USA     | 192.168.246.135 | 1       |
| 14       | sbaelish | 2022-05-10 | 10:20:18   | US      | 192.168.16.99   | 1       |
| 15       | jrafael  | 2022-05-08 | 17:17:26   | USA     | 192.168.168.51  | 0       |

|     |          |            |          |        |                 |   |
|-----|----------|------------|----------|--------|-----------------|---|
| 183 | nmason   | 2022-05-11 | 05:29:36 | CANADA | 192.168.137.147 | 0 |
| 184 | alevitsk | 2022-05-08 | 03:09:48 | CAN    | 192.168.33.70   | 0 |
| 185 | jsoto    | 2022-05-10 | 13:34:58 | USA    | 192.168.151.91  | 0 |
| 186 | bisles   | 2022-05-09 | 04:29:17 | USA    | 192.168.40.72   | 0 |
| 188 | jsoto    | 2022-05-11 | 00:39:09 | USA    | 192.168.21.88   | 0 |
| 189 | nmason   | 2022-05-08 | 05:37:24 | CANADA | 192.168.168.117 | 1 |
| 190 | jsoto    | 2022-05-09 | 05:09:21 | USA    | 192.168.25.60   | 0 |
| 191 | cjackson | 2022-05-08 | 06:46:07 | CANADA | 192.168.7.187   | 0 |
| 192 | bisles   | 2022-05-10 | 08:32:03 | USA    | 192.168.201.40  | 1 |
| 193 | lrodrigu | 2022-05-08 | 07:11:29 | US     | 192.168.125.240 | 0 |
| 194 | jclark   | 2022-05-12 | 14:11:04 | CAN    | 192.168.197.247 | 0 |
| 195 | alevitsk | 2022-05-11 | 06:59:13 | CANADA | 192.168.236.78  | 1 |
| 196 | acook    | 2022-05-10 | 09:56:48 | CAN    | 192.168.52.90   | 0 |
| 197 | jsoto    | 2022-05-08 | 09:05:09 | US     | 192.168.36.21   | 0 |
| 200 | jclark   | 2022-05-12 | 01:11:45 | CANADA | 192.168.91.103  | 1 |

144 rows in set (0.001 sec)

I used the `WHERE` clause and `NOT` operator to filter the outputs and receive the login attempts outside Mexico. However, the word “Mexico” could be “Mex”, “MEX”, and etc. To simplify this, I chose `LIKE` with `MEX%` as the pattern to match as `MEX` and `MEXICO`. The `%` sign indicates any unspecified characters when used with `LIKE`. As a result, there were 144 login attempts outside Mexico.

## Retrieve employees in Marketing

My team wants to update certain computers across departments. I created a SQL query to filter for employee machines from employees in the Marketing department in the East building.

```
MariaDB [organization]> SELECT *
->
-> FROM employees;
```

| employee_id | device_id    | username | department             | office      |
|-------------|--------------|----------|------------------------|-------------|
| 1000        | a320b137c219 | elarson  | Marketing              | East-170    |
| 1001        | b239c825d303 | bmoreno  | Marketing              | Central-276 |
| 1002        | c116d593e558 | tshah    | Human Resources        | North-434   |
| 1003        | d394e816f943 | sgilmore | Finance                | South-153   |
| 1004        | e218f877g788 | eraab    | Human Resources        | South-127   |
| 1005        | f551g340h864 | gesparza | Human Resources        | South-366   |
| 1006        | g329h357i597 | alevitsk | Information Technology | East-320    |
| 1007        | h174i497j413 | wjaffrey | Finance                | North-406   |
| 1008        | i858j583k571 | abernard | Finance                | South-170   |
| 1009        | NULL         | lrodrigu | Sales                  | South-134   |
| 1010        | k242l212m542 | jlansky  | Finance                | South-109   |
| 1011        | l748m120n401 | drozas   | Sales                  | South-292   |

```

MariaDB [organization]> SELECT *
->
-> FROM employees
->
-> WHERE department = 'Marketing' AND office LIKE 'East%';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1000 | a320b137c219 | elarson | Marketing | East-170 |
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |
| 1075 | x573y883z772 | fbautist | Marketing | East-267 |
| 1088 | k865l965m233 | rgosh | Marketing | East-157 |
| 1103 | NULL | randerss | Marketing | East-460 |
| 1156 | a184b775c707 | dellery | Marketing | East-417 |
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |
+-----+-----+-----+-----+-----+
7 rows in set (0.001 sec)

```

I first selected all the data in the `employee` table and used the `WHERE` clause to filter employees who are part of the marketing team and reside in the east building using `AND office LIKE 'East%'`. As a result, there are 7 employees who match the criteria.

## Retrieve employees in Finance or Sales

Across departments, plenty of employee data needs to be updated. I created a SQL query to filter for employee machines from employees in the Finance or Sales departments.

```

MariaDB [organization]> SELECT *
->
-> FROM employees
->
-> WHERE department = 'Finance' OR department = 'Sales';
+-----+-----+-----+-----+-----+
| employee_id | device_id | username | department | office |
+-----+-----+-----+-----+-----+
| 1003 | d394e816f943 | sgilmore | Finance | South-153 |
| 1007 | h174i497j413 | wjaffrey | Finance | North-406 |
| 1008 | i858j583k571 | abernard | Finance | South-170 |
| 1009 | NULL | lrodrigu | Sales | South-134 |
| 1010 | k242l212m542 | jlansky | Finance | South-109 |
| 1011 | l748m120n401 | drosas | Sales | South-292 |
| 1015 | p611q262r945 | jsoto | Finance | North-271 |
| 1017 | r550s824t230 | jclark | Finance | North-188 |
| 1018 | s310t540u653 | abellmas | Finance | North-403 |
| 1022 | w237x430y567 | arusso | Finance | West-465 |
| 1024 | y976z753a267 | iuduike | Sales | South-215 |
| 1025 | z381a365b233 | jhill | Sales | North-115 |
| 1029 | d336e475f676 | ivelasco | Finance | East-156 |
| 1035 | j236k303l245 | bisles | Sales | South-171 |
+-----+-----+-----+-----+-----+

```

|      |              |          |         |             |
|------|--------------|----------|---------|-------------|
| 1147 | r454s225t299 | tvega    | Finance | West-177    |
| 1148 | s328t505u907 | dharvey  | Finance | South-181   |
| 1159 | d881e710f732 | jshen    | Finance | East-193    |
| 1164 | i682j513k442 | fsmeltz  | Finance | North-163   |
| 1169 | NULL         | mmitchel | Sales   | Central-250 |
| 1174 | s371t911u987 | eortiz   | Finance | North-428   |
| 1175 | t959u687v394 | jclark2  | Finance | North-194   |
| 1176 | u849v569w521 | nliu     | Sales   | West-220    |
| 1181 | z803a233b718 | sessa    | Finance | South-207   |
| 1185 | d790e839f461 | revens   | Sales   | North-330   |
| 1186 | e281f433g404 | sacosta  | Sales   | North-460   |
| 1187 | f963g637h851 | bbode    | Finance | East-351    |
| 1188 | g164h566i795 | noshiro  | Finance | West-252    |
| 1195 | n516o853p957 | orainier | Finance | East-346    |

71 rows in set (0.001 sec)

I selected the Finance department and Sales department. By using the `WHERE` clause and `OR` operator I filtered the outputs to make sure all employees who are members of both departments are listed. As a result, there are 71 people who happen to be members of both departments.

## Retrieve all employees not in IT

I created a SQL query to filter for employee machines from employees not in the Information Technology department.

```
MariaDB [organization]> SELECT *
->
-> FROM employees
->
-> WHERE NOT department = 'Information Technology';
```

| employee_id | device_id    | username | department      | office      |
|-------------|--------------|----------|-----------------|-------------|
| 1000        | a320b137c219 | elarson  | Marketing       | East-170    |
| 1001        | b239c825d303 | bmoreno  | Marketing       | Central-276 |
| 1002        | c116d593e558 | tshah    | Human Resources | North-434   |
| 1003        | d394e816f943 | sgilmore | Finance         | South-153   |
| 1004        | e218f877g788 | eraab    | Human Resources | South-127   |
| 1005        | f551g340h864 | gesparza | Human Resources | South-366   |
| 1007        | h174i497j413 | wjaffrey | Finance         | North-406   |
| 1008        | i858j583k571 | abernard | Finance         | South-170   |

```

1180 | y1312211a378 | medwards | Human Resources | Central-340
1181 | z803a233b718 | sessa | Finance | South-207
1183 | b566c710d544 | lquraish | Human Resources | East-400
1184 | c986d200e170 | ptsosie | Human Resources | Central-247
1185 | d790e839f461 | revens | Sales | North-330
1186 | e281f433g404 | sacosta | Sales | North-460
1187 | f963g637h851 | bbode | Finance | East-351
1188 | g164h566i795 | noshiro | Finance | West-252
1189 | h784i120j837 | slefkowi | Human Resources | West-342
1190 | NULL | kcarter | Marketing | Central-270
1191 | NULL | shakimi | Marketing | Central-366
1194 | m340n287o441 | zwarren | Human Resources | West-212
1195 | n516o853p957 | orainier | Finance | East-346
1198 | q308r573s459 | jmartine | Marketing | South-117
1199 | r520s571t459 | areyes | Human Resources | East-100
-----+-----+-----+-----+-----+
161 rows in set (0.001 sec)

```

First, I started by selecting all data from the `employee` table. Then, I used a `WHERE` clause with `NOT` to filter for employees not in the IT department.

## Summary

I applied filters to SQL queries to get specific information on `employee` and `log_in_attempts` tables. I used the `AND`, `OR`, `NOT` operators to filter for the specific information and I used `LIKE` and the `(%)` sign filter for patterns.